

- Ideal OCD Threshold = $15\text{ A} \times 5\text{ m}\Omega = 75\text{ mV}$.
 - However, the closest options are 72 mV (0x0A) and 78 mV (0x0B), providing 14.4 A and 15.6 A, respectively. Both options have the RSNS bit = 1.
 - 0x0A (14.4A) will be used in this example.
- The OCD delay threshold setting for a 320-ms delay is 0x05.
- Therefore, PROTECT2 should be programmed with 0x5B.

备注

Care should be taken when determining the setting of OV_TRIP and UV_TRIP as these are ADC value outputs and correlation to cell voltage also requires consideration of the ADC GAIN and ADC OFFSET registers. More specific details can be found in [节 8.3.1.2](#).

- Configure the Voltage-based protection settings through OV_TRIP, UV_TRIP and PROTECT3:
 - The selected OV Threshold is 4.30 V. If ADCOFFSET is 0 and GAIN is 382, the desired threshold is 11257 or 0x2BF9.
 - Therefore, OV_TRIP should be programmed with 0xBF.
 - The selected UV Threshold is 2.5 V. If ADCOFFSET is 0 and GAIN is 382, the desired threshold is 6545 or 0x1991.
 - Therefore, UV_TRIP should be programmed with 0x99.
 - The selected OV Delay is 2 s and the selected UV Delay is 4 s.
 - Therefore, PROTECT3 should be programmed with 0x50.

9.2.3 Application Curves

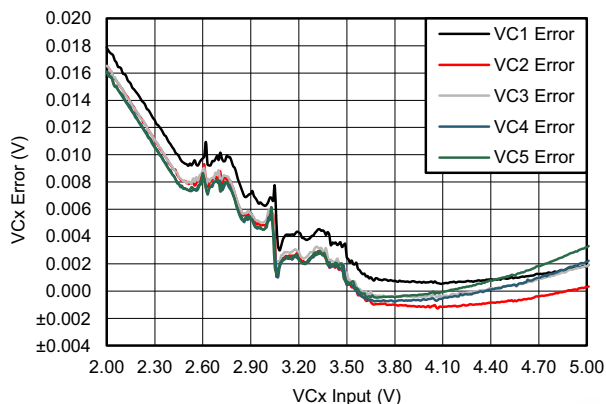


图 9-5. BQ76930 VCx Error Across Input Range at 25°C

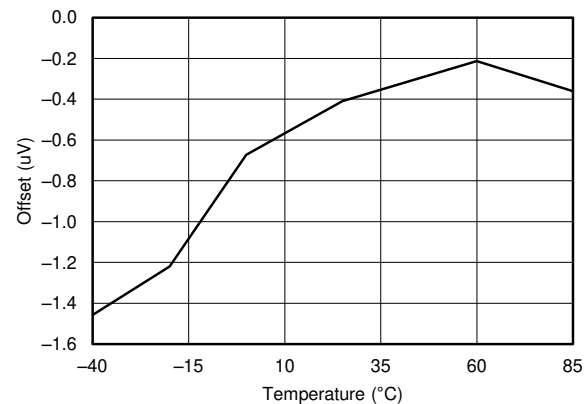


图 9-6. Coulomb Counter Offset

10 Power Supply Recommendations

The BQ769x0 devices are powered through the BAT and REGSRC pins but the BQ76930 and BQ76940 have additional ‘Power’ pins to provide the power to the entire device in the higher cell configurations.

The use of Rf and Cf connected to the BAT pin, noted in the typical application diagrams, are required to filter system transients from disturbing the device power supply. These components should be placed as close as to the IC as possible.

Additionally, for the BQ76930 and BQ76940 there are additional requirements to ensure a stable power supply to the device. The REGSRC pin is powered through a source follower circuit where the FET is used to provide current for REGSRC from the battery positive terminal while reducing the voltage to a suitable value for the IC. The FET also dissipates the power resulting from the load current and dropped voltage external to the IC and care should be taken to ensure the correct dissipation ratings are specified by the chosen FET.

The BQ76920 does not use a FET because the battery voltage is within the REGSRC range.